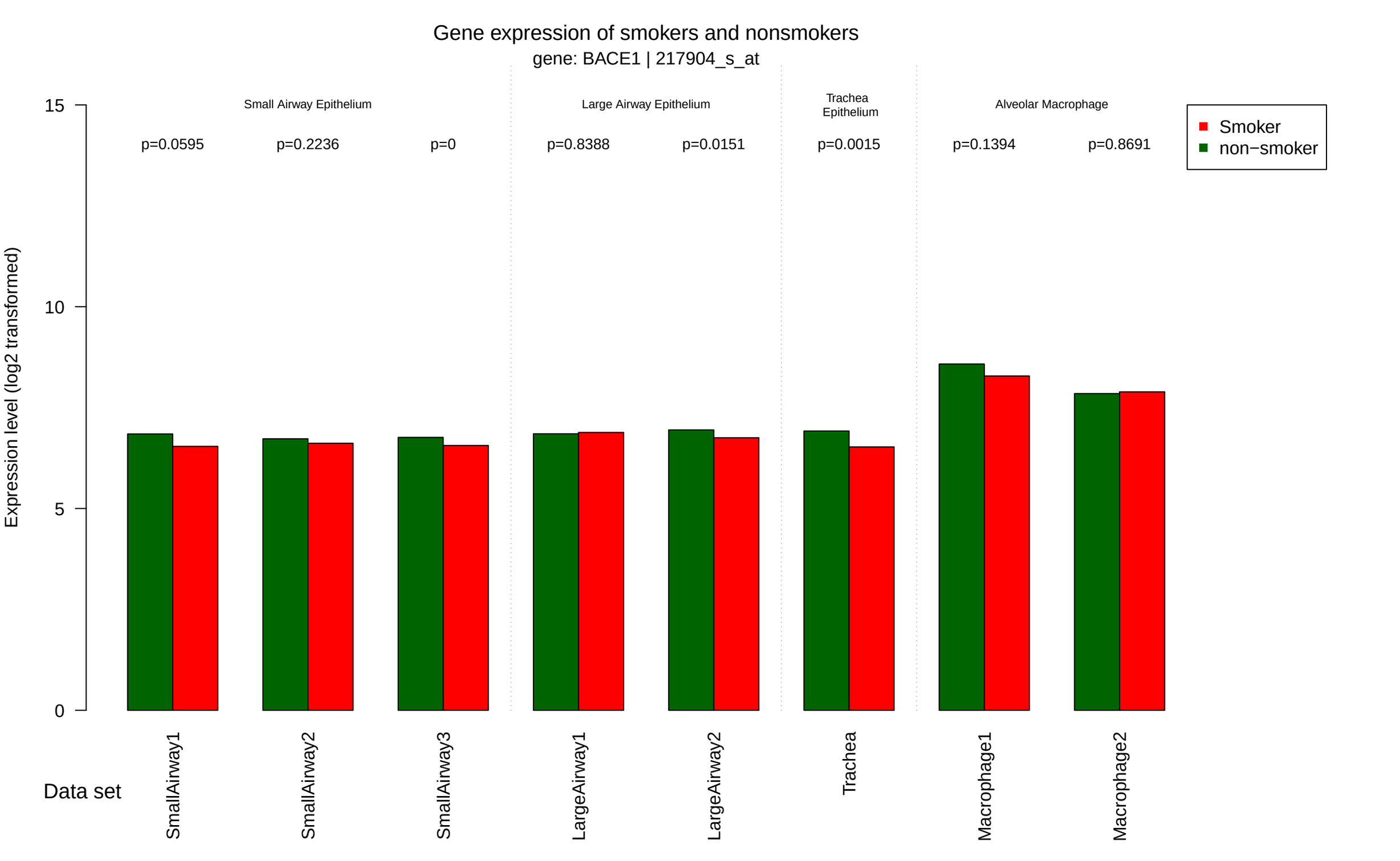
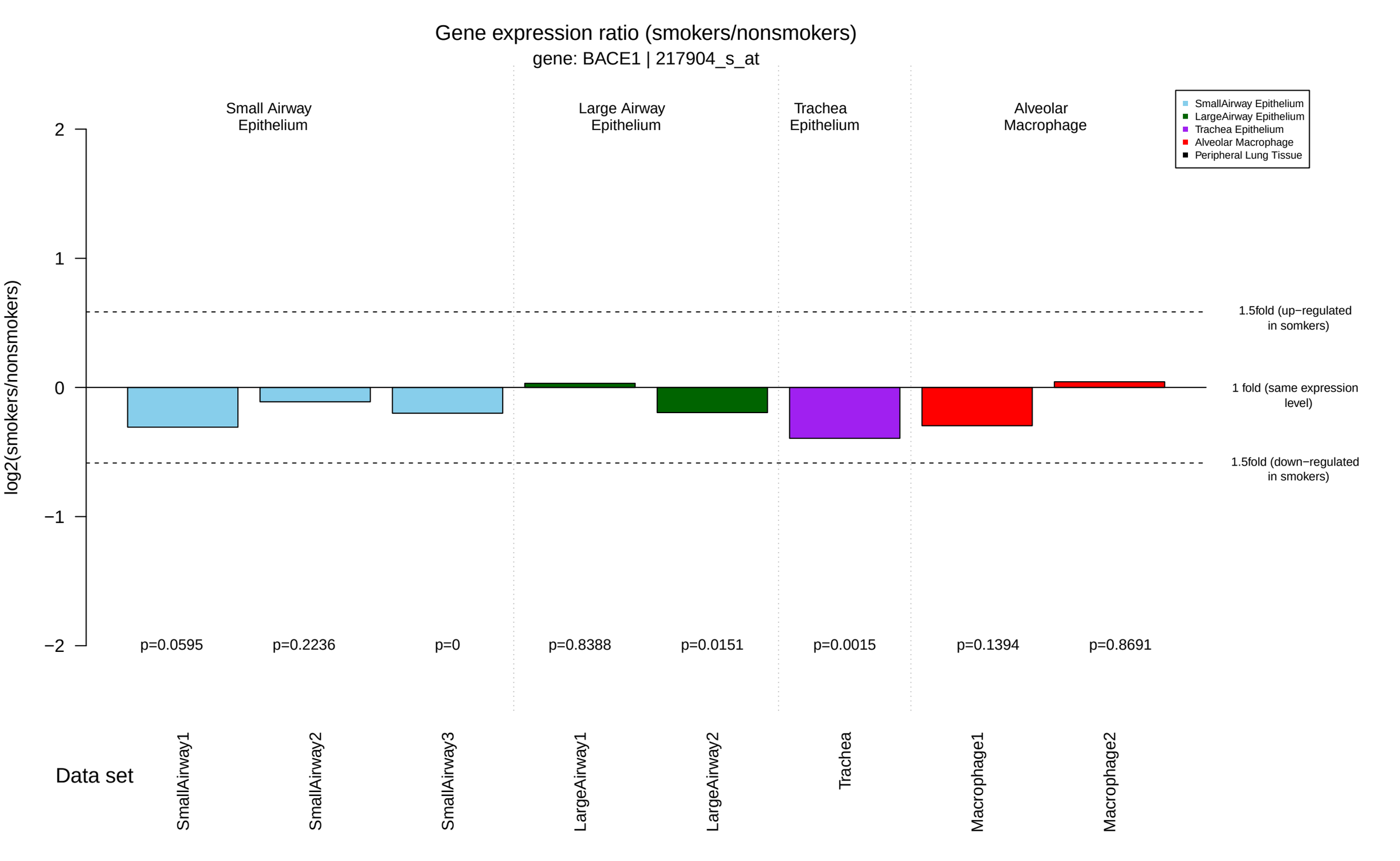
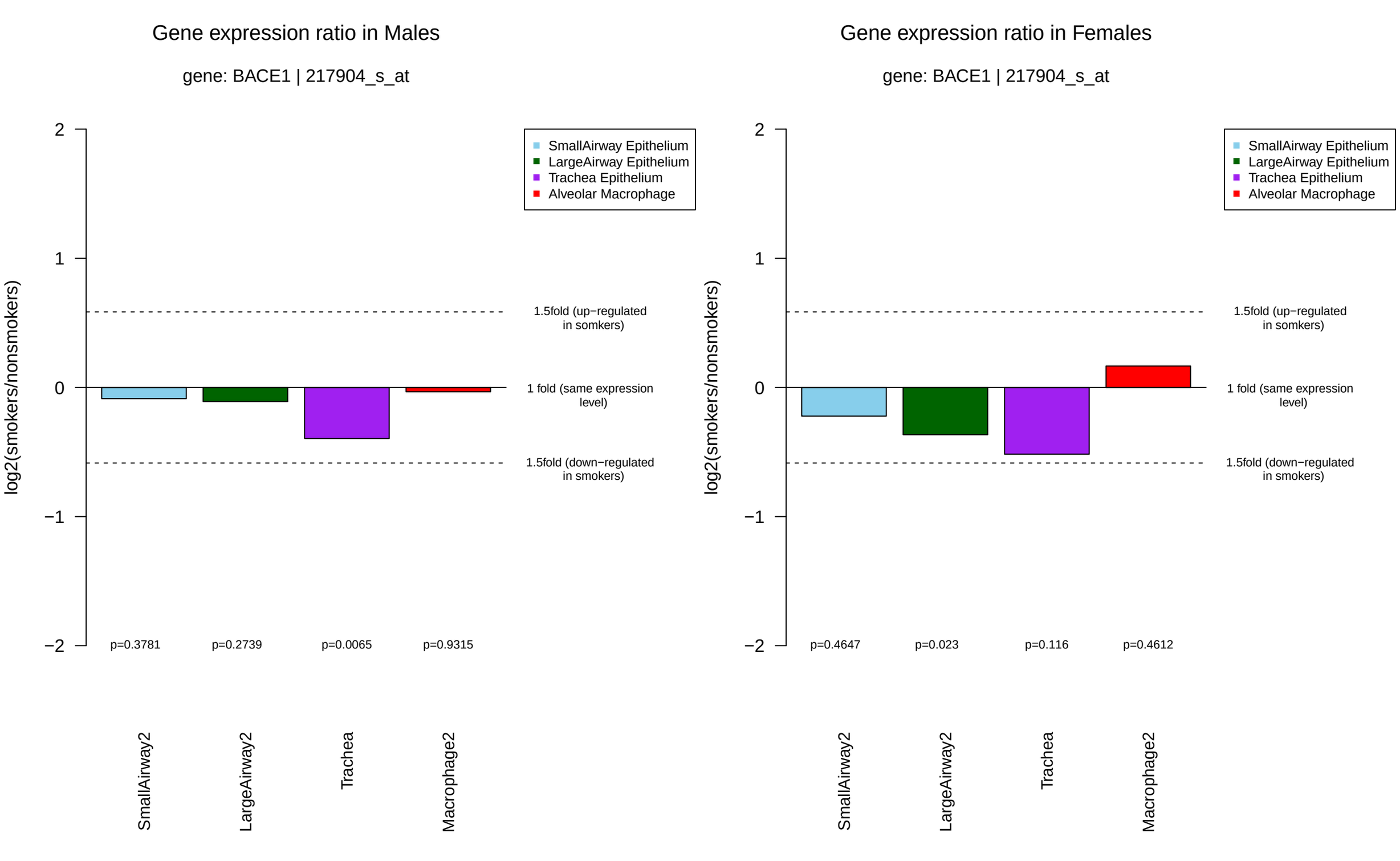
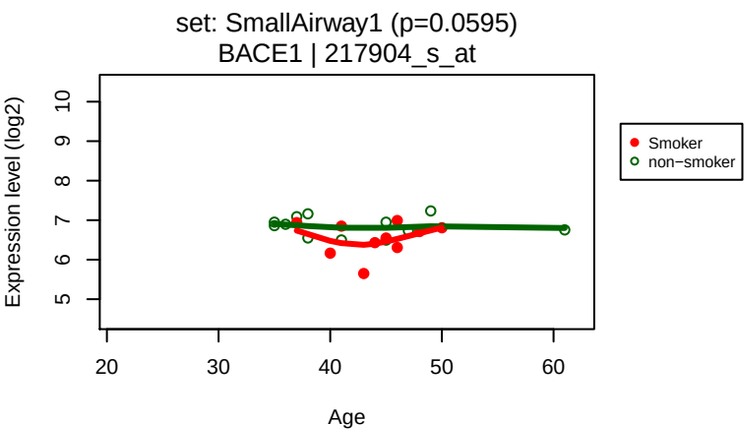
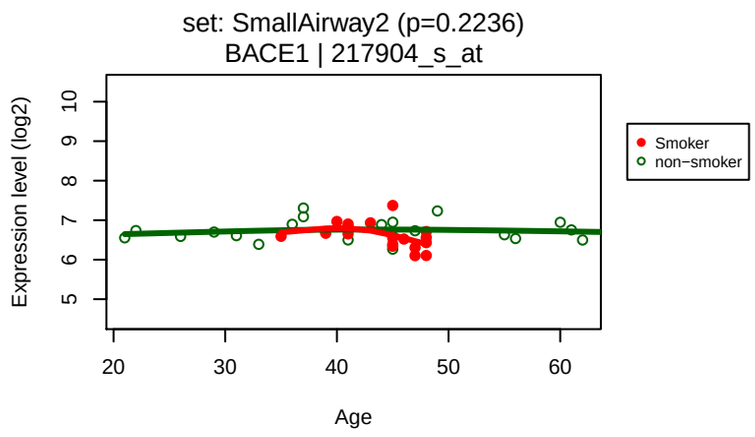
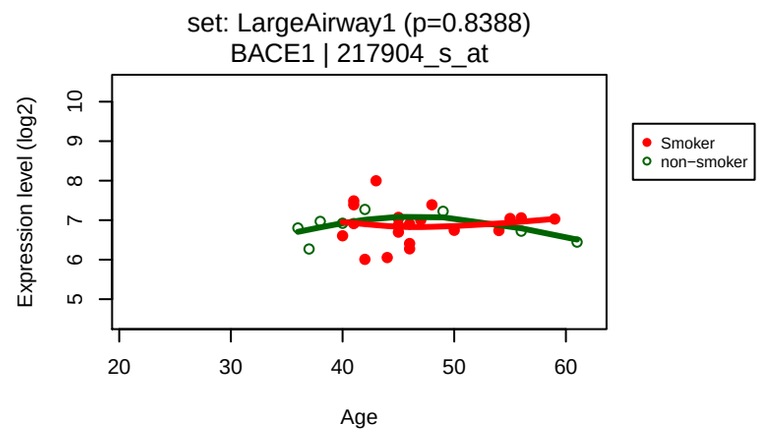

Gene expression of smokers and nonsmokers
gene: BACE1 | 217904_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0595
p=0.2236
p=0
p=0.8388
p=0.0151
p=0.0015
p=0.1394
p=0.8691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

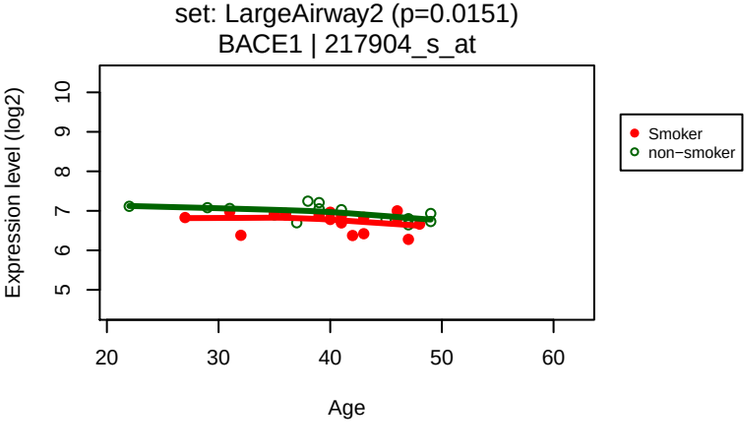

Gene expression ratio (smokers/nonsmokers)
gene: BACE1 | 217904_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0595
p=0.2236
p=0
p=0.8388
p=0.0151
p=0.0015
p=0.1394
p=0.8691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

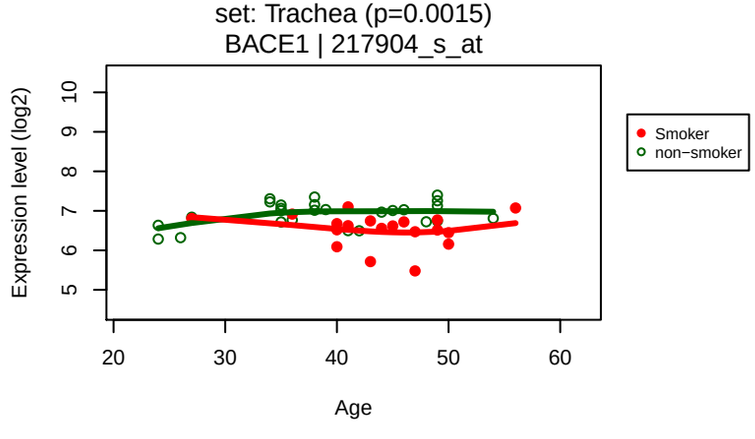

Gene expression ratio in Males
gene: BACE1 | 217904_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3781
p=0.2739
p=0.0065
p=0.9315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACE1 | 217904_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4647
p=0.023
p=0.116
p=0.4612
SmallAirway2
LargeAirway2
Trachea
Macrophage2

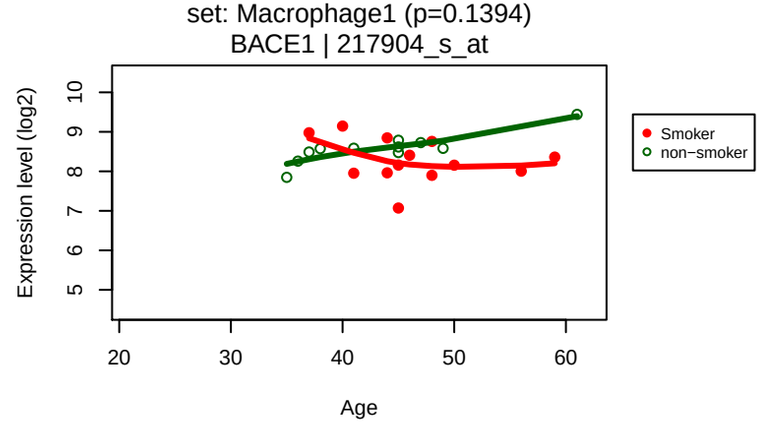

set: SmallAirway1 (p=0.0595)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker

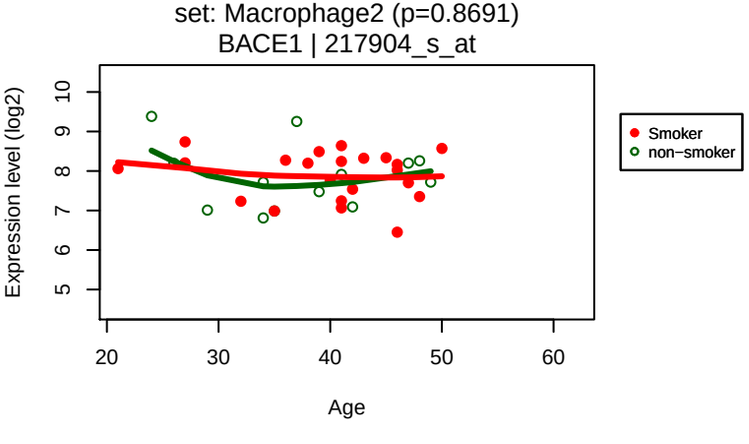

set: SmallAirway2 (p=0.2236)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker

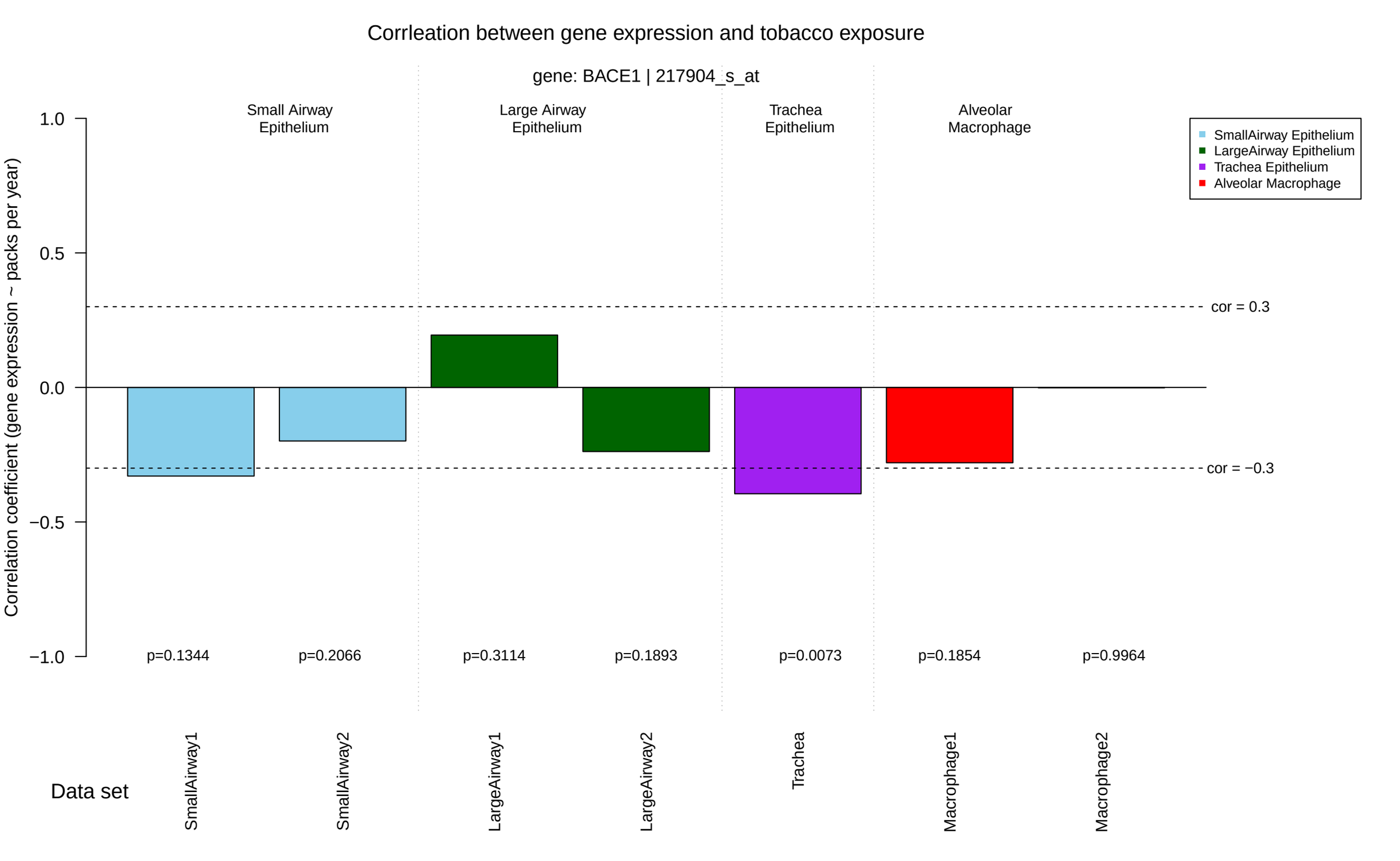

set: LargeAirway1 (p=0.8388)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0151)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0015)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1394)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8691)
BACE1 | 217904_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACE1 | 217904_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1344
p=0.2066
p=0.3114
p=0.1893
p=0.0073
p=0.1854
p=0.9964
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2